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Verilog Implementation of Biometric-Based Transmission of Fused Images Using Data Encryption Standards Algorithm



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Abstract Biometric scan images are used for authentication purposes widely. Because they are unique in nature, the use of biometric scan images is increasing day-by-day in authentication and security. Due to the economic-based systems such as banking and financial sectors are using biometric-based security and authentication, the use of biometric images is increasing every day. In this paper, the fingerprint of the user is taken as key image for DES algorithm and it is used for the encryption of fused image. Hardware implementation of data encryption standards algorithm is designed. For enhancing the target information content in image, using MATLAB software, two multi-focus images are fused. The dual-tree-complex wavelet transform is used for the fusion of images. The comparison is done with other hardware designs in terms of number of LUTs and power and complexity of implementation with other encryption techniques. This paper gives the secure transmission of fused images using DES encryption, with the use of biometric image as a key image, which is strong and unique method for security and authentication in nowadays.

Keywords Image fusion · Wavelet transform · Data encryption standards · Encryption · Decryption · S-box · Verilog hardware description language

1 Introduction

Biometric scan images such as fingerprint, face, iris, thumb, and hand impression are used for authentication and security purposes widely. Because they are distinctive in nature, its use is increasing day-by-day in authentication and security. Due to the

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economic-based systems such as banking and financial sectors are using biometric-based security and authentication of the use of biometric images is increasing every day. We use dual-tree complex wavelet transform (DT-CWT) here to fuse two multi-focus images.

There are various fusion techniques available today, and their utilization based on requirement is chosen by user. Here in this paper, DT-CWT is used because of its quality as compared to other fusion techniques like averaging method, Pyramid method, curvelet, DWT, and SWT. After the DT-CWT-based fusion operation, DES encryption is applied to this fused image, using the biometric scan image of user such as fingerprint as a key image. Recovery of securely transmitted image is possible by using the same authorized key image [1, 2]. Otherwise, the original fused image is not obtained and this is the main aim of this work done in this paper. Verilog implementation of DES using Xilinx ISE tool is done; design of the hardware and other parameters, like timing, power, and number of LUTs, slices, are obtained. The comparison is done with other hardware designs in terms of number of LUTs and power and complexity of implementation with other encryption techniques like triple DES, AES, and Blowfish [3]. Here the goal of this paper is to provide biometric-based secure transmission of images using encryption technique, as well as to reduce complexity of algorithm used in encryption in terms of key size and block size [4], which are the parameters affecting the area and power distribution and the ease of implementation. Biometric-based secure transmission is strong and unique method for security and authentication in nowadays.

1.1 Related Work

The various fusion techniques have been used by other authors such as averaging pixels, maximum, PCA, and pyramid decompositions and transform-based techniques. Elbirt (et al.) gives the complete details about image fusion and its principles, methods, and applications, and has evaluated FPGA implementation and performance [2].

The details of image fusion techniques based on wavelets are provided by Krista Amolims (et al.). For the segmentation of input image features, a dual-tree complex wavelet transform is used, either jointly or separately to create a map of region [4].

New updated version of the AES-based image encryption algorithm is proposed by Kamali (et al.) [6].

The image encryption algorithm was proposed using the chaotic map and S-DES by Bin and Liu (et al.) [7]. Agarwal and Bedi suggested the use of hybrid image fusion technique to enhance the function of clinical diagnosis [11].

2 Preliminaries

DWT-based approaches have now largely replaced previous fusion methods based on decompositions of pyramid [4]. It is possible to explain the fusion of two test images using the discrete wavelet transform in terms of the 2 recorded input test images S_0 and S_1, the WT is **ω,** and the fusion rule is φ, which is taken as symbol to combine colocated coefficients within the domain of transform. And coefficients of the fused wavelet are inversed to generate the resulting fused image IF using inverse wavelet transform $\boldsymbol{\omega - 1}$ hence [4]

$$\mathbf{IF} = \boldsymbol{\omega} - \mathbf{1}(\varphi(\boldsymbol{\omega}(\mathbf{S0}), \boldsymbol{\omega}(\mathbf{S1}))).$$

2.1 Image Fusion Based on Dual-Tree Complex Wavelet Transform (DT-CWT)

While image fusion based on DWT offers nice results, the associated shift variance has been recognized to produce sub-optimal performance, which is because of the magnitude of a coefficient does not adequately reflect the actual energy due to position of spatial frequency. Hence, it is not a reliable base for comparing perceptive significance inside the domain of transform [3, 4]. Using transforms such as shift-invariant discrete wavelet transform (SI-DWT) has addressed the problem [12]. The transformation is similar to the DWT, but at each point of the decomposition, it eliminates down-sampling. Since down-sampling allows the variance of the shift, the SI-DWT is shift-invariant.

This differentiation (of + and − frequencies) leads to splitting the original three DWT sub-bands into 6 DT-CWT sub-bands based on the orientations of +15/−15, +45/− 45, and +75/− 75, respectively.

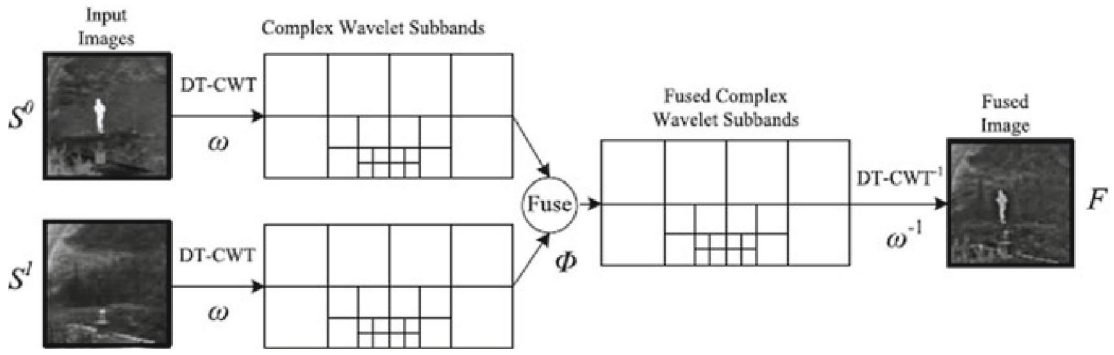


Fig. 1 Image fusion of test images using DT-CWT [15]

The sub-band coefficients resulting are complex, and the magnitude of each coefficient is about the invariant of changes. Figure 1 displays a typical fusion scenario.

Figure 1 shows that it is possible to create and transform coefficients from test images. The fused transformed is achieved. The most popular fusion rule is for the coefficient of wavelets which is max rule of fusion [13].

Input : Multi-focus Images: I_1, I_2

$W^0 \leftarrow$DT-CWT(I_1)

$W^1 \leftarrow$DT-CWT(I_2)

$W^2 \leftarrow$Max-Coefficient Fusion(W^0,W^1)

Initial fused DT-CWT transform

F=DT-CWT^{-1}(W^2)

Output: Fused image I_F

The selection of max rule of fusion or its variants presupposes that a coefficient's perceptual value is directly related to its magnitude [14, 15].

Test images were first processed using DT-CWT for getting transformed coefficients. Theses coefficients are further fused using fusion rule. The inverse DT-CWT is then applied over fused coefficients to get fused image F.

2.2 Data Encryption Standards

This is a block cipher of archetypal-type. This algorithm accepts the input of 64-bit block size and converts it into ciphertext of same length.

DES also uses a key to customize the encryption in such a way that only those who know the particular key used to encrypt are able to decrypt. The secret key consists theoretically of 64 bits; however, the algorithm actually only uses 56 of those bits. Figure 2 displays the entire DES algorithm [1, 3, 5].

A. **S-box**

DES S-boxes differ for all other permutations. [9] The structure of S-box design is shown at the end of Fig. 3.

B. **Verilog implementation for S-box**

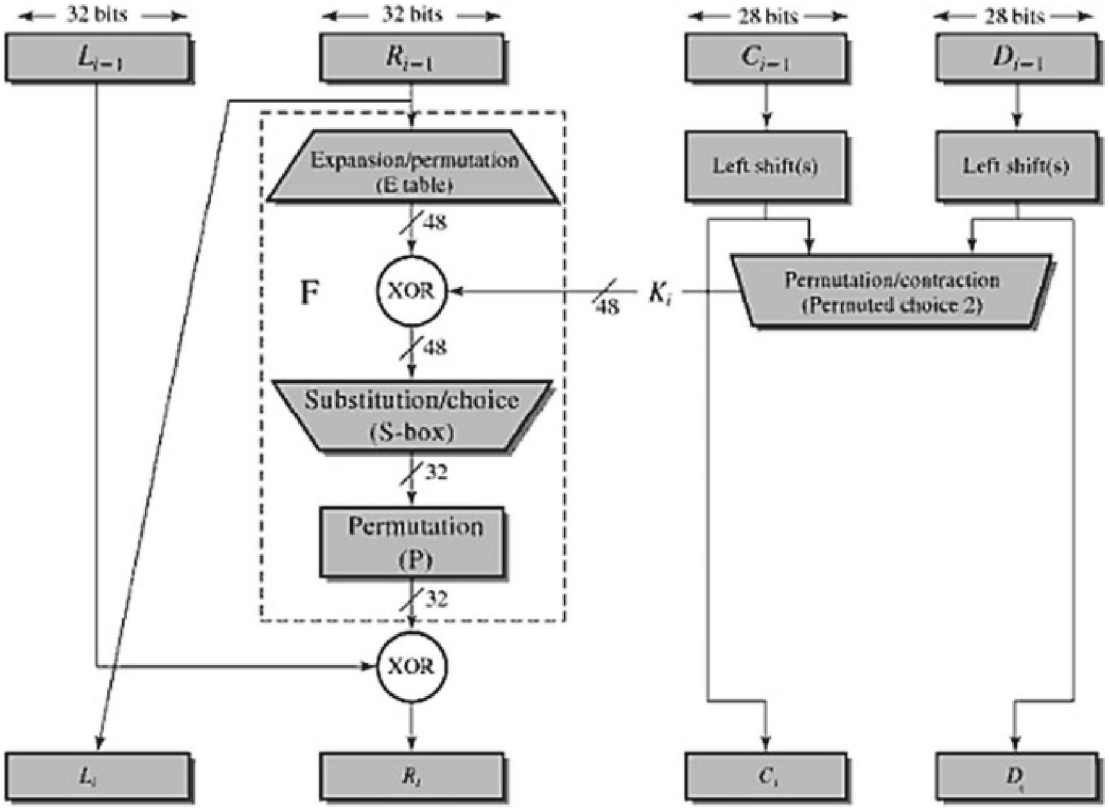


Fig. 2 DES algorithm [1]

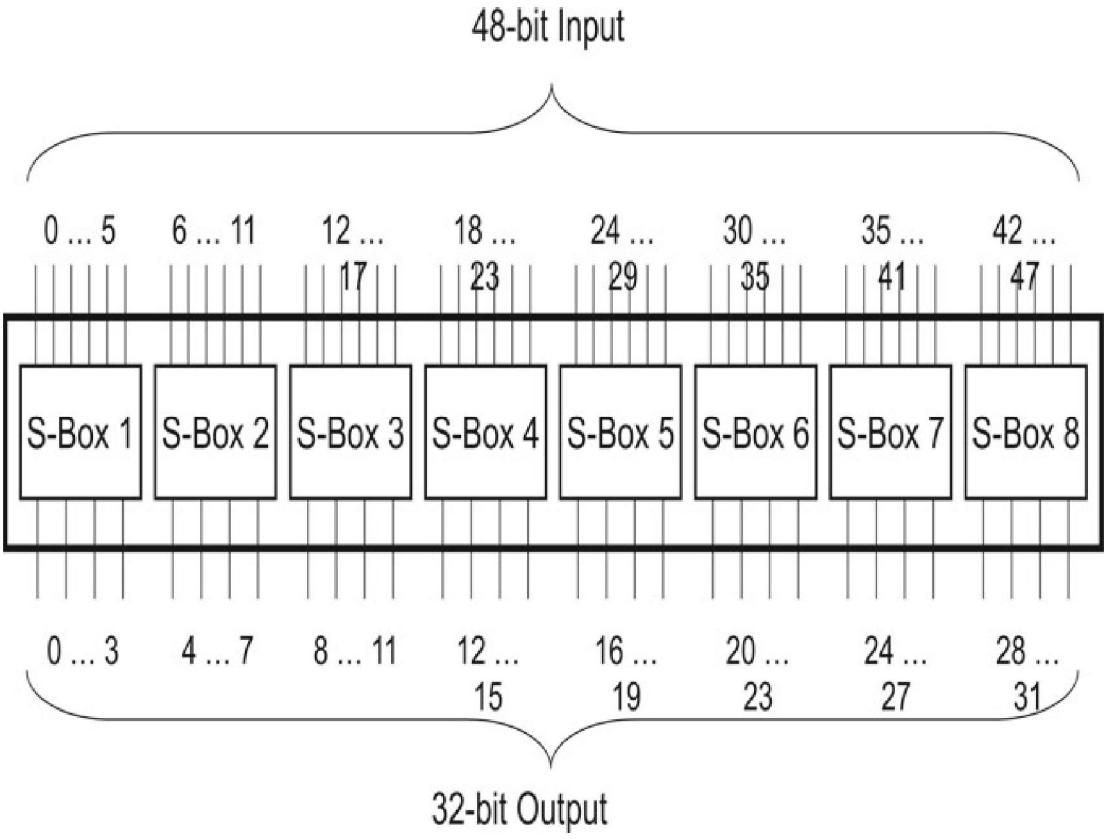


Fig. 3 Eight DES S-boxes [7]

B. Verilog implementation for S-box

```
module Sbox1(in, out);
input [1:6] in;
output [1:4] out;
reg [1:4] out;
always @(in) begin
  case ({in[1], in[6], in[2:5]}) //synopsys full_case
parallel_case
0: out = 14;
1: out = 4;
2: out = 13;
3: out = 1;
4: out = 2;
5: out = 15;
6: out = 11;
7: out = 8;
.
63: 0ut= 13;
  end case
  end
module
```

Similarly we apply it for the other 7 s-boxes.

3 Proposed Architecture

The whole sequence of this work is organized into three parts. The first part is presented using the block diagram shown in Fig. 4. In this part, we performed DT-CWT on two test images, which results in fused image I_F [16, 17]. DT-CWT-based fused image I_F is encrypted using DES algorithm. The unique biometric scan image like fingerprint, iris, or hand impression is used as a key image I_K in order to secure transmission of fused image and also in storage which is shown in Fig. 5. The same key image (fingerprint) is used for DES decryption process, when the user needs his original fused image. After DES decryption process, the original fused image is recovered as shown in Fig. 6. If the key image used is not authorized, then the original fused image is not recovered after decryption. This ensures the secure transmission.

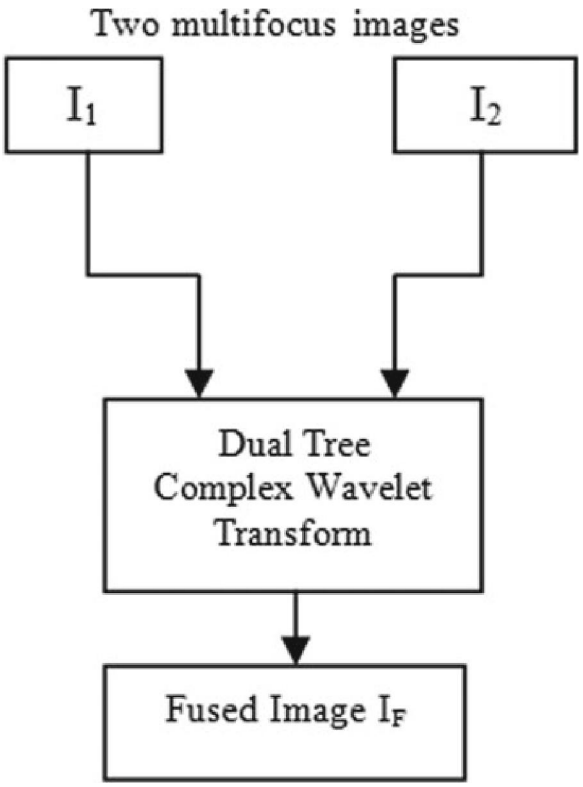


Fig. 4 DT-CWT-based image fusion

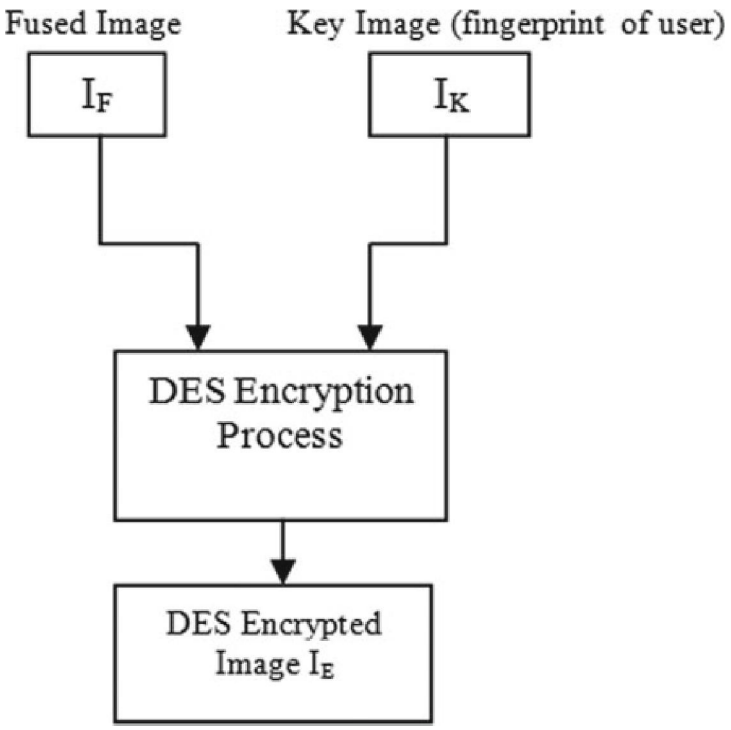


Fig. 5 DES algorithm

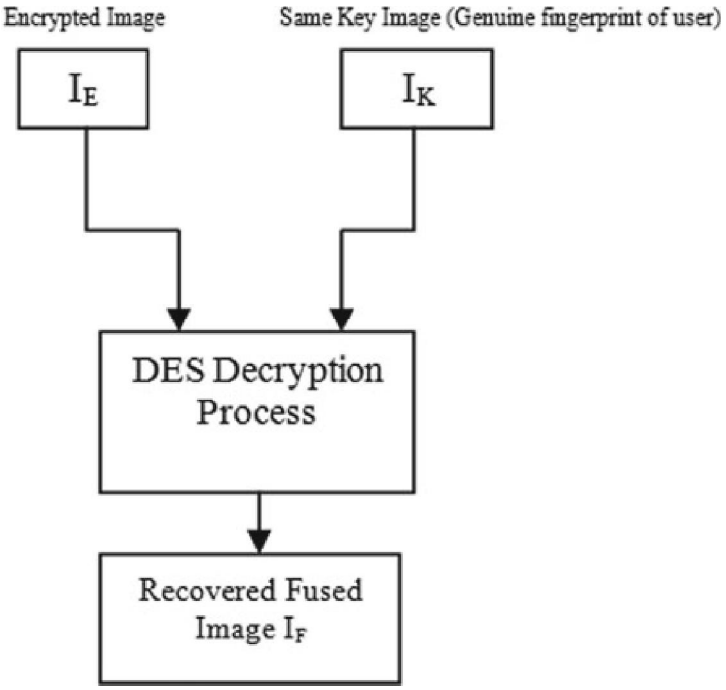
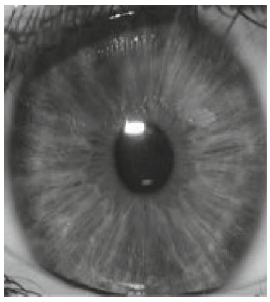


Fig. 6 DES decryption



Biometric images Iris



Fingerprint



Hand impression [11]

4 Assessment Environment

In this paper, we use MATLAB and Xilinx programming. We now use the VLSI format here. Hence, hardware specification language is used to create this design is Verilog HDL. Similarly, requirements and specialties of Verilog make popular for different performance and speed of design implementation. Xilinx ISE 14.2 Design Suit is used as simulation software as well as for synthesis layout [8, 12]. Likewise, software and hardware are used and it is simple to implement on them.

5 Implementation Results

The first result is found using DT-CWT of two multi-focus test images as shown in Fig. 7 using MATLAB, and hence, fused image is found as indicated in Fig. 8. After this, fused image as proposed is encrypted using fingerprint of user with DES method in Xilinx ISE 14.2 Verilog implementation.

These two test images are taken, and they are fused with DT-CWT method in MATLAB of the fused image is stored, and in the proposed scheme, it is encrypted with other key image which is the biometric data of the user and it may be his fingerprint, iris, thumb impression, or hand impression. Both fused image and key image are encrypted using DES method. DES is implemented on Xilinx ISE 14.2. The proposed results are shown in Fig. 9.

After DES, we get encryption result of fusion image and biometric image and then it is used for secure transmission and storage. After decryption using the same unique fingerprint image as a key, we get fused image again. Figure 10 shows the implementation on Xilinx ISE 14.2. The proposed scheme is tested for single block using DES. The input data stream is 64-bit single block, and the key is 56-bit single block as per DES operation. Using this entire programming environment, we get fused image with more information from test images. Further, we get chip-level solution using Verilog HDL implementation for security with DES algorithm.



Image1 Image2

Fig. 7 Two multi-focus test images [13]



Fig. 8 Fused image I_F

Figure 9 shows the proposed results for Verilog implementation of biometric-based transmission of fused images using DES.

Analysis of Power Distribution in Chip:

The power distribution analysis is essential in the design because the leakage power and dynamic power of the designed chip are also accountable in optimum design that is shown in Table 1. It gives the distribution of power consumed in clocks, logics, signals, and inputs/outputs (IOs).



a) Fused Image I_F b) Fingerprint c) DES Encrypted image d) Decrypted image

Fig. 9 Proposed scheme

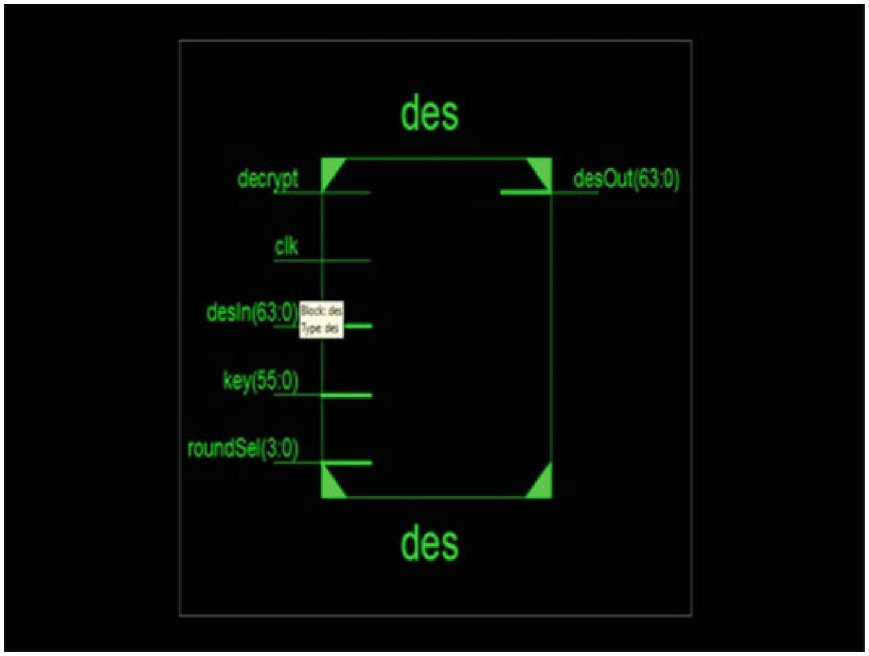


Fig. 10 Xilinx ISE 14.2 implementation

Table 1 Power distribution in chip

On chip	Power(W)	Used
Clocks	0.002	1
Logic	0.003	437
Signals	0.008	673
IOs	0.131	190
Leakage	0.041	–
Total	0.184	–

Table 2 Comparison table for power distribution

Design	Power (W)
Sumanth Kumar Reddy S et al.	0.655
Ours	0.184

Table 3 Encryption algorithms

Algorithm	Key Size (Bits)	Block Size (Bits)
DES	64	64
3DES	192	64
AES (Rijndael)	256	128
Blowfish	448	64

The power distribution window is generated by clicking appropriate option of *analyze power distribution* in the Xilinx ISE 14.2

6 Comparisons

Here the comparison is made with the other author's results such as Sumanth Kumar Reddy et al. [10]. Table 2 shows the power distribution comparison.

Further Table 3 shows the generally used other algorithm such as triple-Data Encryption Standards (DES), Advanced Encryption Standards (AES) (Rijndael), and Blowfish. Also we see in this DES algorithm with respect to number of bits used, the key size and block size are minimum as used in other algorithms.

Therefore, it is obvious from the above comparison that this design comparison in proposed scheme in terms of power distribution gives better results as compared to other work done. Also, it utilizes a fix 64-bit key and allows 64-bit block size, and therefore, making it easier to implement on Verilog HDL in Xilinx ISE14.2 tools.

7 Conclusions

Each person has a unique identity, and their biometric image is stored with their records in a database. The DES key which is a biometric data is not accessible to unauthorized user. So it becomes more secure. Same key is used for decryption of transmitted fused images. Only authenticated users can access their data after giving their fingerprint or iris scan genuine information, etc. Hence, we conclude that image fusion is very useful in increasing the information content of target image and DES algorithm performed secure transmission of fused image.

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